

304 – OOP & DS Lab Programs

Week – 1

1. Display various patterns using cout and cin. (Minimum 5 different patterns)
2. Write a program to print Fibonacci series up to given terms.
3. Write a program to check whether the given number is perfect or not. [123→ 1+2+3 =6 and 1*2*3=6]
4. Write a program to find out and print all prime numbers between given range.

Week – 2

5. Create the menu driven program to make the calculator. The option of the menus are:
 1. Addition
 2. Substraction
 3. Multiplication
 4. Division
 5. Average
 6. Maximum
 7. Minimum
 8. Exit
6. Write a program to increment the value of variable using the concept of call by value and call by reference.
7. Write a program to find middle number from three numbers using the concept of return by reference.
8. Write a program to find maximum of three numbers using user defined functions
9. Write a program to illustrate the use of scope resolution operator (::).

Week – 3

10. Write a program to create a class name student having data members rollno, name and three marks and member functions like,
 - a. Input rollno, name and marks.
 - b. Calculate percentage.
 - c. Display all information.
11. Write a program to input five records of person class which have members name, age and member functions for inputting data and display information of person whose age is maximum.
12. Write a program of inline function for finding maximum and minimum from two numbers.
13. Write a program to find area of circle by using the value of PI as default argument.

Week – 4

14. Design two classes to support following type conversin:
Rupee R; Dollar D;
D=R; converts Rupee to Dollar. Write a program to carry out this conversion.
15. Write a program to implement single inheritance.
16. Write a program to implement multilevel inheritance.
17. Write a program to implement multiple inheritance.

Week – 5

18. Write a program to implement hierarchical inheritance.
19. Write a program to implement hybrid inheritance.

20. Create a class publish which have members like publisher id and publisher name. Create one another class book with members like author name, book name and price which is inherited from class publish. Write a program to display information of both these classes.
21. Create a class called “Vehicle” which contains data member’s registration number and fuel type. Make get_data() function to input data value. Create class “two_wheeler” from “Vehicle” which contains data members distance and mileage. Make getdata() function to input data. Use overriding techniques for getdata() function and display the information with fuel used.

Week – 6

22. Create class “SYBCA” containing the data member’s name of student, sear number, marks in CPP, marks in DS, and marks in DBMS. Take the appropriate member function and get data into class until user’s choice. Display the information of all students having the marks more than 180.
23. Class Customer contains Customer_id, Name, Address as data member and add_customer and display_customer as member function. A derived class Order is created from the class Customer with publicly inherited. The derived class contains Price1, Price2, and Price3 as prices of 3 products and input_prices and display_result as member functions. Create an array of object of the Order class and display the result of 5 customers.
24. Write a program to find area of rectangle, triangle and circle through function overloading
25. Write a program to add two time objects and display the sum using a friend function.
26. Write a program to add two distance objects (kilometre, meter) and display the sum using a friend function.

Week – 7

27. Write a program to find the average of two entered complex numbers and display the average using a friend function to class.
28. Write a program to overload + and – binary operators.
29. Write a program to overload < and > to compare two strings.
30. Write a program to overload assignment operator (=).

Week – 8

31. Write a program to implement virtual function.
32. Create a class publish which have members like publisher id and publisher name. Create one another class book with members like author name, book name, and price which is inherited from class publish. Write a program to display information of both the classes.
33. Write a program to implement stack with push, pop, and peep operation.

Week – 9

34. Write a program to reverse the string using stack.
35. Write a recursive program for Tower of Hanoi.
36. Write a program to convert infix into postfix expression using stack.

Week – 10

37. Write a program to convert infix into prefix expression using stack.
38. Write a program to evaluate postfix expression.
39. Write a program to evaluate prefix expression.

Week – 11

40. Write a recursive program for finding factorial.
41. Write a recursive program for Fibonacci series.
42. Write a menu driven program to implement simple queue with all operation.
43. Write a menu driven program to implement circular queue with all operation.

Week – 12

44. Write a menu driven program to implement D-queue with all operation.
45. Write a menu driven program to implement output restricted D-queue with all operation.
46. Write a menu driven program to implement input restricted D-queue with all operation.